

Saliency-Guided Prompt Compression and Document-Level Source Selection with SigExt for Abstractive Summarization

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Abstract—Large Language Models (LLMs) achieve strong performance in abstractive summarization, but prompting with full documents is costly at scale and could exceed context-window limits. We leverage SigExt, which provides a Longformer-based saliency extractor that assigns importance scores to keyphrases, to design prompt compression strategies with the aim of retaining highly informative content under a fixed token budget. In this work, we propose two extensions.

(1) On CNN/DailyMail, we implement both sentence-level and phrase-level compressors driven by SigExt keyphrase saliency. The results show that phrase-level compression reduces input tokens by more than 70% while preserving ROUGE and BERTScore.

(2) On Multi-News, we introduce document-level ranking by aggregating keyphrase saliency to identify the most informative sources in each cluster of documents. In our experiments, keeping only the top two documents reduces the input size by approximately 28% while producing only a minimal decrease in the summary quality.

Overall, our results indicate that SigExt saliency scores can be translated into practical and optimal compression strategies that reduce cost and latency while maintaining the quality of the generated summary.

I. INTRODUCTION

The field of LLM abstractive summarization has achieved significant results in recent years. However, generating high-quality summaries starting from the full documents as prompts could be very expensive at scale in terms of costs and latency, and this might be even a crucial challenge when the LLM context window is limited. A natural mitigation is to shorten the prompt while preserving the most relevant information to generate the summary and preserve its quality.

SigExt [1] is a Longformer-based extractor that assigns an importance score to each *keyphrase* identified in the source document. In our implementation, these phrase-level scores are the core signal used to decide what to keep or discard when building shorter prompts for downstream LLMs. Intuitively, SigExt promotes phrases that are central to the document meaning and discourages less relevant details; this yields a lightweight, model-agnostic representation of what matters that can be used to power different prompt-construction policies. Moreover, the output is a ranked list of keyphrases with scores, which makes the compression step transparent: changing the budget simply shifts a threshold on the score ranking and therefore it does not require a retraining stage.

We study two settings, each with a dedicated solution.

(i) **Prompt compression.** We propose two SigExt-driven

prompt compressors. The *phrase-level* compressor builds the prompt selecting the top ranked keyphrases and linking them, trying to minimize the length of the prompt. The *sentence-level* compressor instead selects whole sentences to preserve the syntactic connections: it assigns to each sentence a score given by the maximum score among the contained keyphrases and then keeps the top ranked sentences under a threshold budget.

We evaluate both approaches at multiple compression ratios and compare GPT-3.5-Turbo and Mistral-7B-Instruct-v0.3, to measure how the summary quality degrades as the input becomes more compressed and less syntactically structured. In both cases, SigExt is used as a primary scoring stage, while the two compressors differ in how aggressively they trade syntax for shortness.

(ii) **Multi-document prompt selection.** We introduce a *document-ranking* strategy to handle the multi-document setting. For each document, we aggregate its SigExt keyphrase scores into a single score of relevance, and then we select the top- k documents (e.g., $k = 2$) and summarize only this cluster. We evaluate this strategy with GPT-3.5-Turbo and Llama-3.1-8B to understand how document selection affects quality and robustness across model sizes. With this discriminative selection, we try to improve prompt efficiency but also to reduce redundancy across source documents.

The source code is available in the [GitHub](#) repository.

II. METHODOLOGY

This section details the SigExt model and our two extensions. We first describe how we use SigExt, then our compression strategies and evaluation pipeline.

A. SigExt

We train SigExt and use it to assign saliency scores to keyphrases. We run this extraction step on both CNN/DailyMail and Multi-News datasets, storing the resulting scored keyphrases; these scores are then used by our prompt-compression and document-selection policies. Following the SigExt pipeline, each document is first processed to extract all keyphrases that are then scored by the Longformer encoder. The saliency score can be interpreted as a signal of relevance for the summarization: important phrases in terms of saliency are more likely to appear in a faithful summary. In our

experiments, we keep the extractor fixed and use it only as a preprocessing module, so that our analysis is focused only on prompt compression and selection approaches.

B. Extension 1: Single-Document Compression

Extension 1 proposes two SigExt-powered prompt-compression strategies that look at different granularities.

1) *Sentence-Level Compression*: The sentence-level compressor maintains the grammatical structure of the whole sentence. We score each sentence s_j by max-pooling the score of the keyphrases it contains,

$$\text{score}(s_j) = \max_{k_i \in s_j} \text{score}(k_i), \quad (1)$$

and keep only the top-ranked sentences under a target keep ratio r . The selected sentences are then concatenated in the original positional order. This policy allows maintaining discourse nuances and limits the risk of generating hallucinated summaries when the downstream LLM is sensitive to missed syntactical connections. However, because the unit of selection is a full sentence, token savings are less fine-grained and can be limited by long and partially less salient sentences.

2) *Phrase-Level Compression*: The phrase-level compressor directly selects a subset of the SigExt keyphrases with respect to a keep ratio r and combines them, maintaining the positional order that they have in the original document, to form the compressed prompt for the LLM. This approach allows a much more granular control of the prompt budget, discarding all the irrelevant keyphrases that are instead maintained at sentence-level, and can capture salient facts even when they are scattered across otherwise unimportant sentences. The main drawback of this strategy is that the resulting prompt is highly unstructured, or in other words, looks like a bag of keyphrases. This can have an important effect on relatively smaller models that rely on syntactic structure to maintain global coherence.

Overall, we could say that sentence-level compression tends to preserve fluency but offers less granular compression, whereas phrase-level maximizes token reduction at the cost of a syntactically unstructured prompt.

3) *Compression Evaluation*: We benchmark both strategies with respect to multiple keep ratios r , chosen to target specific prompt lengths.

C. Extension 2: Multi-Document Selection

Extension 2 focuses on the multi-document summarization setting. We choose the Multi-News dataset, which contains clusters of documents, each cluster on the same news and probably with overlapping or redundant content. Generally, the whole cluster is provided as a prompt to the LLM. Instead, the multi-document selector extracts only a small, highly informative subset of documents in each cluster and then provides it as a prompt to reduce the prompt length compared to the baseline case.

In particular, each Multi-News sample consists of multiple documents $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_m\}$ concatenated with a delimiter. We score each document by aggregating the scores of the

keyphrases that lie within its span, and then select the top- k documents to form the shorter version of the prompt.

- 1) *Segmentation and phrase assignment*: We split the input into its source documents and compute the character span of each document. We then assign each keyphrase to the document whose span contains it.
- 2) *Score aggregation*: For each document $\mathcal{D}_i$, we aggregate the salience of its assigned keyphrases. Since SigExt returns log-probabilities, we convert them to non-log scores via exponentiation before aggregation:

$$\text{raw_score}(\mathcal{D}_i) = \sum_{k_j \in \mathcal{D}_i} \exp(\text{score}(k_j)) \quad (2)$$

- 3) *Length normalization*: We normalize by the number of keyphrases assigned in each document, since longer documents tend to contain more keyphrases than shorter ones. This could introduce a bias, causing the selector to penalize shorter documents:

$$\text{score}(\mathcal{D}_i) = \frac{\text{raw_score}(\mathcal{D}_i)}{|\{k_j : k_j \in \mathcal{D}_i\}|}. \quad (3)$$

- 4) *Top- k selection*: We rank documents by normalized score and keep the top- k . The selected documents are linked in their original order to form the reduced prompt.

To summarize, we can say that this process implements a two-layer framework, which first scores documents by the aggregated and normalized importance, then summarizes only the top- k units.

D. LLM Inference Pipeline

1) *Model Selection*: We evaluate both extensions with multiple LLMs to inspect how a larger proprietary model and smaller open-weight alternatives behave differently with respect to different compression/selection methods:

- GPT-3.5-Turbo (OpenAI API): the main API-based baseline used throughout the experiments. [2]
- Mistral-7B-Instruct-v0.3 (Extension 1): a local 7B open-weight model used to test sensitivity to compressed prompts, in particular, compared to the phrase-level scenario; deployed with 4-bit quantization. [3]
- Llama-3.1-8B-Instant (Extension 2): an 8B model accessed via Groq for low-latency inference in the multi-document setting. [4]

To ensure reproducibility, we use deterministic generation settings across all models (temperature 0, maximum 512 new tokens, greedy decoding). For GPT-3.5 we also fix the seed to 42; for Mistral, we use 4-bit quantization to fit on commodity GPUs.

2) *Prompt Engineering*: We use zero-shot prompting with XML-tagged outputs for reliable parsing:

```
Here is a news article:
<text>
Please write a short summary for the
article in 1-2 sentences.
Write your summary in <summary> XML tags.
```

For multi-document scenarios (Extension 2), the prompt is adapted:

```
Here are some news articles:
<text>
Please write a detailed and comprehensive
summary of the provided articles. The
summary should cover all key points and
relevant details.
Write your summary in <summary> XML tags.
```

E. Evaluation Metrics

1) *ROUGE Scores*: We compute ROUGE-1, ROUGE-2, and ROUGE-L using the standard implementation [5]. ROUGE measures n-gram overlap precision (P), recall (R), and F1 between generated and reference summaries. We report only the F1 scores in the results tables.

2) *BERTScore*: BERTScore [6] evaluates semantic similarity using contextual embeddings from BERT. We use the rescaled baseline configuration with the English model, reporting F1 scores. BERTScore is particularly valuable for assessing whether compressed inputs preserve semantic content despite lexical changes.

III. RESULTS

A. Extension 1: Single-Document Compression

1) *Experimental Results*: We evaluate Extension 1 on 200 CNN/DailyMail samples, measuring how summary quality changes as we reduce the prompt length. The keep ratios r are selected to guarantee an input reduction in the 60–80% range (from $\sim 61.6\%$ up to 79.3%), enabling a direct comparison between token reductions and the corresponding drop in evaluation metrics. Tables I and II report the full-text baseline and SigExt-based compression results for both sentence- and phrase-level prompts. We report ROUGE (F1) and BERTScore (F1).

a) *GPT-3.5-Turbo*: GPT-3.5 exhibits a favorable efficiency–quality trade-off: across the full 60–80% reduction range, ROUGE varies only slightly. With the highest compression (79.3% reduction), ROUGE-1 stays close to the baseline for both sentence prompts (37.92 vs. 39.33) and phrase prompts (38.90 vs. 39.33), showing that SigExt concentrates enough signal to preserve the core content even when the prompt is cut to roughly one fifth of its original size.

As compression becomes milder (down to $\sim 61.6\text{--}61.7\%$), the scores slightly exceed the full-text baseline (sentence $r=0.33$: R-1 39.72, R-2 15.00; baseline R-2 14.96), suggesting that moderate pruning can remove less important context (e.g., redundant details) without harming coverage. BERTScore remains comparatively stable (within about 1.2 points of the baseline across configurations), indicating that semantic coverage is almost maintained.

b) *Mistral-7B*: For Mistral, the same 60–80% prompt reduction produces a larger and more consistent degradation, revealing a tighter prompt-length budget before quality drops. At 79.3% reduction, sentence prompts fall from R-1 37.15 to 34.70 (and BERTScore from 23.70 to 21.85), showing that removing too much context impairs both overlap and semantics. Phrase-level compression at the same reduction yields a higher ROUGE-1 (36.17) but the lowest BERTScore (21.41), which suggests that the model can match some salient n-grams from the keyphrase prompt while still losing global coherence/semantic alignment. When the reduction is relaxed to $\sim 61.6\text{--}61.7\%$, performance recovers substantially (sentence $r=0.33$: R-1 36.99; BERTScore 23.46), implying that for smaller models the token savings are best targeted around the lower end of our range (near 60%), where prompt shortening is meaningful but enough discourse structure remains to support generation.

c) *Trade-off at matched reductions*: At matched reductions (e.g., $\sim 79\%$), GPT-3.5 shows only minor metric changes, making high compression viable when cost/latency are the priority. Conversely, Mistral incurs a clearer quality penalty at the same token savings, and benefits more from sentence-level prompts as compression becomes milder. Overall, the experiments quantify a simple guideline: the larger the reduction target (toward 80%), the more the choice of prompt structure matters, and the more smaller models benefit from retaining sentence-level context.

B. Extension 2: Multi-Document Selection

1) *Experimental Results*: We evaluate Extension 2 on 200 Multi-News samples and quantify the trade-off between prompt length and summary quality; we compare FULL TEXT prompting with our top- k selection framework for $k \in \{1, 2\}$. Table III reports the average prompt reduction together with ROUGE (F1) and BERTScore (F1) for GPT-3.5-Turbo and Llama-3.1-8B.

TABLE I
EXTENSION 1 (CNN/DAILYMAIL): PROMPT COMPRESSION RESULTS
(GPT-3.5; ROUGE, BERTSCORE).

Config.	Input Red.	R-1	R-2	R-L	BERT
FULL TEXT	0.0%	39.33	14.96	25.02	27.67
Sentence $r=0.18$	79.3%	37.92	13.99	24.16	26.52
Sentence $r=0.24$	71.8%	38.65	14.37	24.85	27.32
Sentence $r=0.33$	61.7%	39.72	15.00	25.65	27.94
Phrase $r=0.30$	79.3%	38.90	13.73	24.12	26.91
Phrase $r=0.40$	73.2%	39.04	13.55	24.16	27.03
Phrase $r=0.60$	61.6%	38.84	13.76	24.31	27.04

TABLE II
EXTENSION 1 (CNN/DAILYMAIL): PROMPT COMPRESSION RESULTS
(MISTRAL-7B; ROUGE, BERTSCORE).

Config.	Input Red.	R-1	R-2	R-L	BERT
FULL TEXT	0.0%	37.15	13.15	23.42	23.70
Sentence $r=0.18$	79.3%	34.70	11.84	22.05	21.85
Sentence $r=0.24$	71.8%	35.86	11.98	22.26	22.94
Sentence $r=0.33$	61.7%	36.99	12.70	22.53	23.46
Phrase $r=0.30$	79.3%	36.17	11.19	21.48	21.41
Phrase $r=0.40$	73.2%	35.71	11.70	21.59	21.71
Phrase $r=0.60$	61.6%	35.92	11.63	21.51	21.84

TABLE III
EXTENSION 2 (MULTI-NEWS): MULTI-DOCUMENT SELECTION RESULTS
(ROUGE, BERTSCORE).

Model	Strategy	Input Red.	R-1	R-2	R-L	BERT
<i>GPT-3.5-Turbo</i>						
Full Text		0.00%	40.73	11.59	19.63	18.40
Top-2		28.06%	40.55	11.47	19.75	17.86
Top-1		64.16%	37.66	10.38	18.84	15.32
<i>Llama-3.1-8B-Instant</i>						
Full Text		0.00%	39.15	12.77	18.61	12.01
Top-2		28.06%	38.68	12.44	18.34	11.02
Top-1		64.16%	36.95	10.58	17.62	7.56

a) Top-2: robust reduction with limited metric loss:

TOP-2 achieves a 28.06% prompt reduction while maintaining near-baseline ROUGE for both models. For GPT-3.5, the drop is negligible (R-1: 40.73 $\rightarrow$ 40.55; R-2: 11.59 $\rightarrow$ 11.47; R-L: 19.63 $\rightarrow$ 19.75), and BERTScore decreases only slightly (18.40 $\rightarrow$ 17.86). This indicates that, in many clusters, two high-salience sources capture most of the cross-document evidence, while removing redundant documents reduces the prompt without materially harming content coverage.

b) Top-1: aggressive reduction and coverage loss: TOP-1 pushes the reduction to 64.16% and leads to a clearer quality drop. For GPT-3.5, ROUGE-1 decreases by 3.07 points (40.73 $\rightarrow$ 37.66) and BERTScore by 3.08 points (18.40 $\rightarrow$ 15.32), suggesting that a single document is often insufficient to cover complementary facts distributed across the cluster. The degradation is stronger for Llama-3.1-8B, which sees a large BERTScore decrease (12.01 $\rightarrow$ 7.56), consistent with a higher dependence on multi-source context for semantic alignment.

Overall, document-level salience ranking enables substantial prompt savings in the multi-document setting, but the reduction budget is critical. $k = 2$ is sufficient to remove a significant amount of redundant content while preserving cluster coverage, while selecting only a single document cuts approximately two thirds of the prompt but at the same time discards too much information that is spread across the group and leads to a sharper quality drop.

IV. CONCLUSION

We presented two practical extensions of SigExt salience scores aimed at improving the efficiency of LLM-based abstractive summarization under prompt-length constraints. Across both tasks, the main takeaway is that keyphrase salience can be converted into concrete prompt-building policies that reduce cost and latency while keeping summary quality largely stable, up to a model- and budget-dependent threshold.

For Extension 1, we showed that salience-guided sentence-level and phrase-level policies can substantially reduce the input while preserving most of the downstream performance. The two approaches trade off different objectives. The phrase-level compressor maximizes information density and yields the largest token savings (often above 70%), but it produces prompts that are less grammatical and may require stronger

instruction-following and robustness from the summarizer. The sentence-level compressor, instead, retains discourse structure and local coherence; while it typically saves fewer tokens, it offers a more reliable quality–budget trade-off for smaller or more sensitive models. Taken together, these results suggest that the optimal compression granularity should be selected not only based on a target token budget, but also on the expected brittleness of the target LLM to fragmented inputs.

For Extension 2 (multi-document selection), we introduced a document-ranking strategy that aggregates keyphrase salience into a document-level relevance score and selects only the top- k sources in each cluster. Experiments on Multi-News indicate that selecting the top-2 documents can remove a meaningful portion of redundant context with near-baseline ROUGE, whereas pushing the reduction to top-1 discards complementary evidence and leads to a clearer coverage loss.

Overall, the results highlight salience extraction as a practical and transparent tool for prompt construction, enabling explicit cost–quality trade-offs. As future work, we plan to compare against prompt-reduction baselines such as LLMingua [7] and Selective Context [8] for Extension 1, and to explore other scoring/selection objectives in document selection for Extension 2.

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